Exploratory Data Analysis Report

1. Project Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on retail stock and sales datasets. The project includes SQL-style analysis, data preprocessing, visualization, and extraction of business insights to understand sales performance, customer behavior, and product trends.

2. Dataset Overview

Datasets used:

- - sales.csv: Contains transaction-level sales data.
- - StockDetails.csv: Contains stock item descriptions.

3. Data Preprocessing and Validation

Sales Dataset Missing Values:

- InvoiceNo: 0
- StockCode: 0
- Description: 0
- Quantity: 0
- InvoiceDate: 0
- UnitPrice: 0
- CustomerID: 0
- Country: 0
- Invoice_Day: 0
- Month: 0
- Hour: 0
- TotalPrice: 0

Stock Dataset Missing Values:

- StockCode: 0
- Description: 0

Sales Dataset Duplicate Rows: 15

Stock Dataset Duplicate Rows: 0

4. Feature Engineering

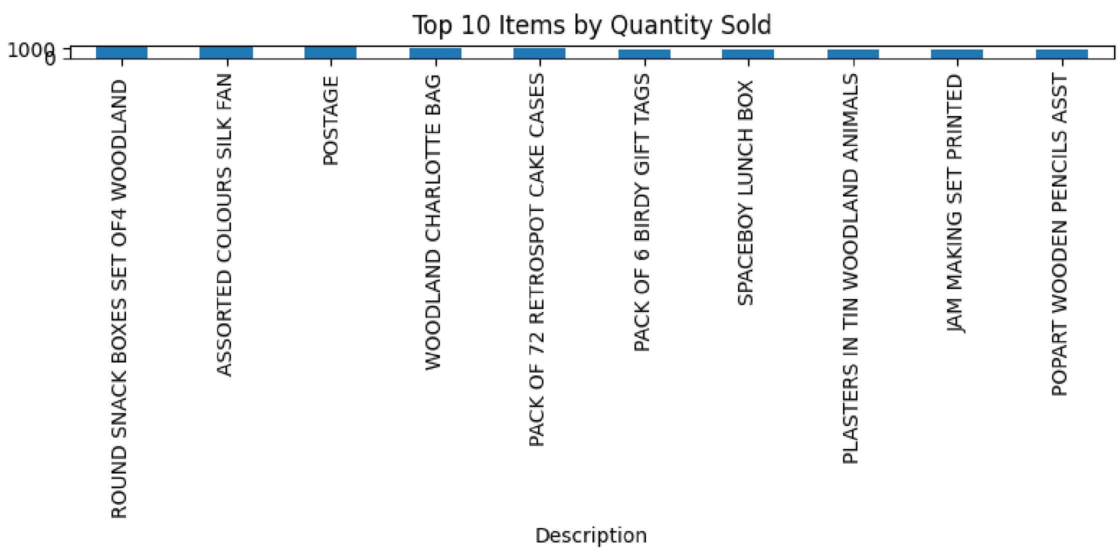
- Converted InvoiceDate into datetime format.
- Extracted Invoice Day, Month, and Hour of transaction.
- Created TotalPrice = Quantity × UnitPrice.

5. SQL and EDA Tasks Summary

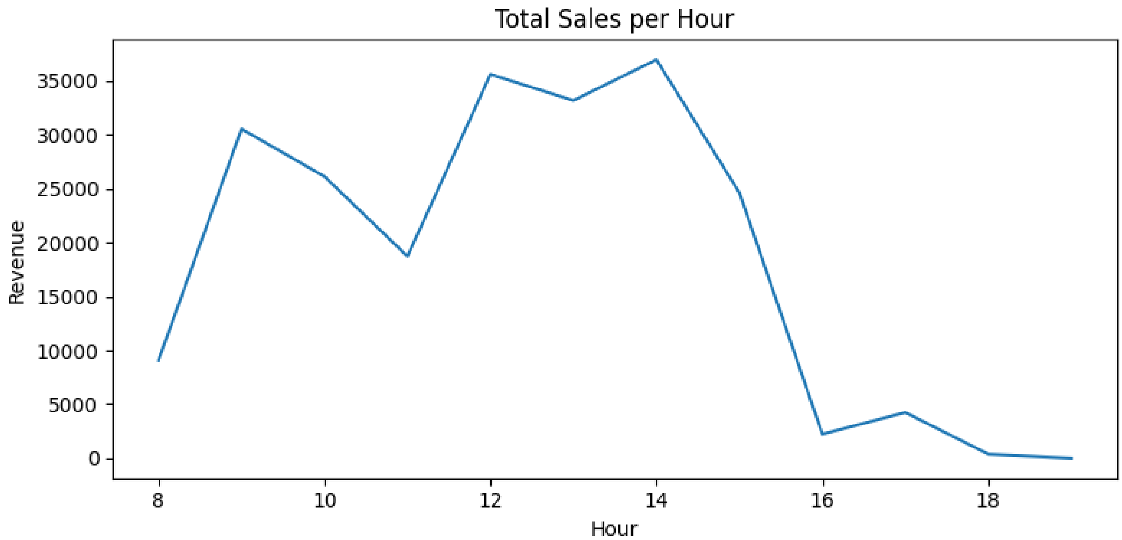
- Retrieved all stock items containing "T-LIGHT" in description.
- Calculated total quantity sold per StockCode.
- Computed total revenue per CustomerID.
- Generated invoice-level distinct stock item counts.
- Performed join between sales and stock details datasets.
- Identified top bestselling products.
- Computed number of unique customers and average quantity per invoice.

6. Visualizations

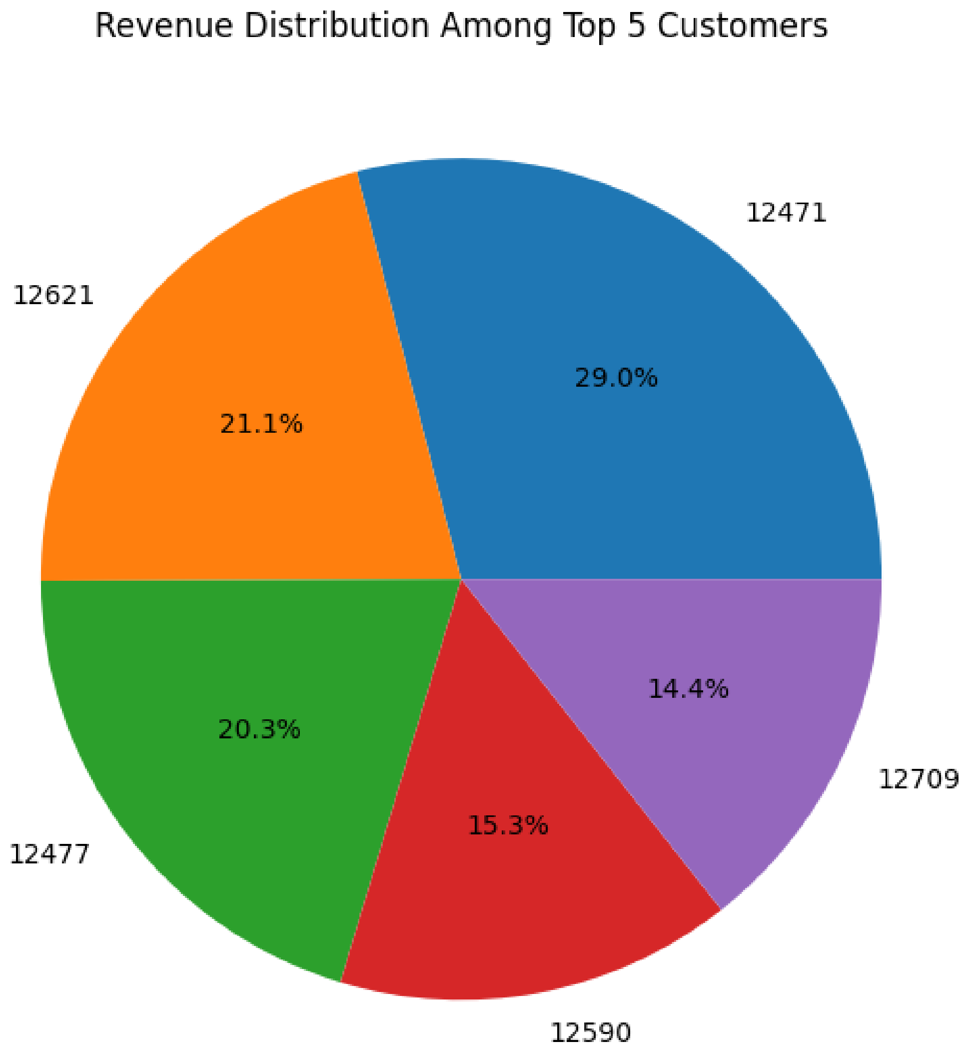
Top 10 Items by Quantity Sold



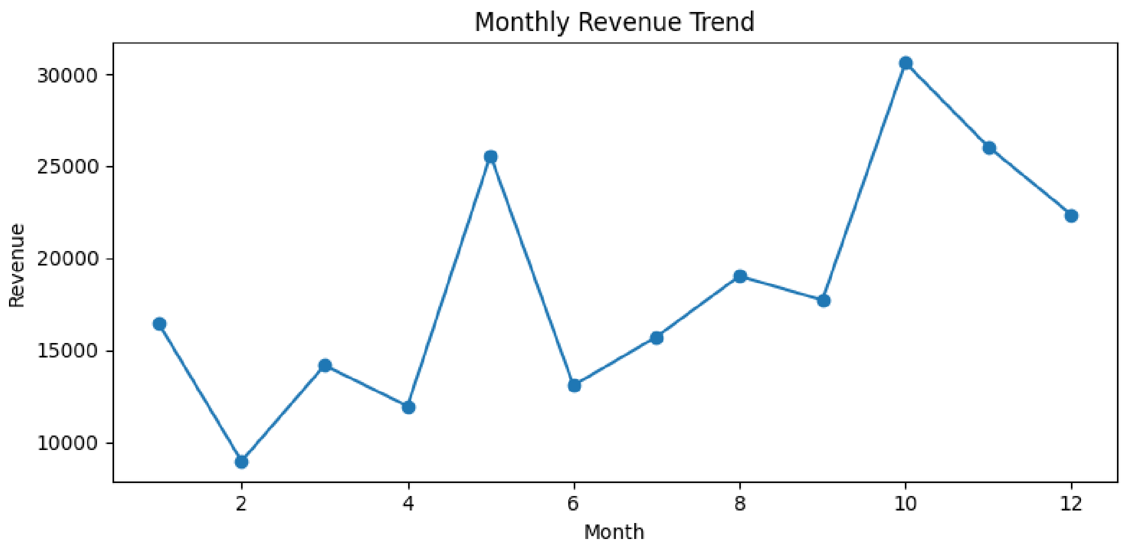
Total Sales per Hour



Revenue Distribution Among Top 5 Customers



Monthly Revenue Trend



7. Business Insights

Highest Revenue Generating Product: POSTAGE with revenue of 20821.00.

Average Order Value per Invoice: 367.66

Customer with Highest Transactions: 12471 (49 invoices)

Number of products in sales data without matching stock details: 0

8. Key Takeaways

- Sales activity peaks during specific shopping hours.
- A small set of products contribute significantly to total sales.
- Top customers account for a substantial portion of revenue.
- Monthly revenue trends help identify seasonality patterns.
- Data preprocessing and feature engineering enabled meaningful business analysis.

9. Conclusion

The project successfully implemented SQL-style analysis, EDA, visualization, and business insight generation using retail stock and sales datasets. The analysis provided valuable understanding of customer behavior, product performance, and revenue trends.